Project Portfolio

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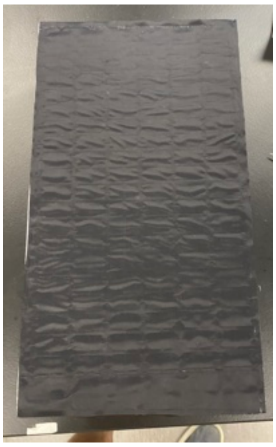
Pouch Motor Manufacturing Machine

In soft robotics, pouch motors are linear actuators that are controlled with pressurised air. One type the FREE lab studies is made by pressing two plastic sheets together with a heated stamp. This manufacturing process is time consuming and low yield, and the lab needs a lot of these pouch motors for demos and experiments. I was hired to make a machine that would automate this production process. The machine was to take in two sheets of plastic, pull them through a set of turning, heated, cylindrical stamps, and spool up the finished product on a roll. I decided to start with a UV DTF laminator, which gave us a frame, heating system, and powered rollers to build on. This was faster than, and similarly priced to, developing our own setup for this generic part of the machine. I then added features that we needed that the laminator did not come with, including the custom roller, which was the most involved part of the project. The stamped roller was made with an aluminium core, and a 3D printed outer shell made of high-temperature SLA resin. We were originally going to make an all aluminium roller, but I suggested trying high temp resin so that I could iterate and iron out kinks in the design before spending most of the project's budget on a part that we weren't sure would work. This let us make daily design changes, come up with simpler solutions to many of the issues we faced, and saved money by only needing to order one machined part.



Uninflated and inflated pouch motors (above and above right)

Laminator before and after modification (above and right respectively)

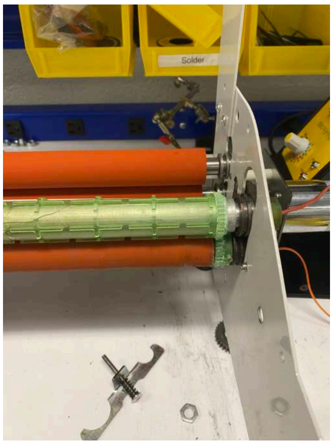
Aluminium core roller with high temp resin shell.

Some of the challenges I encountered with the project were thermal expansion, rotational alignment of the top and bottom rollers, and an uneven pressure distribution between individual stamp elements.

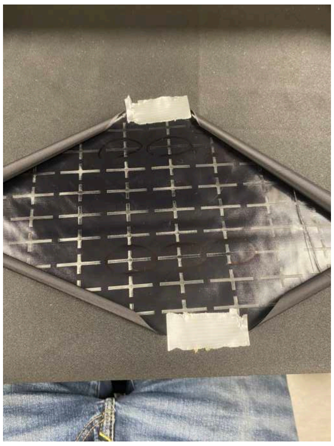
Thermal expansion: the high temp resin had a lower coefficient of thermal expansion than aluminium, resulting in internal stresses that led to cracking after multiple heat cool cycles. I resolved this by slightly oversizing the resin print, so that it would fit the aluminium part properly at the operating temperature instead of room temperature. Another expansion issue was that the rods heat from an element at their centre, so there were significant temperature differences between the resin near the centre and on the outer edge, also leading to cracking. I reduced this stress by making the outer resin shell as thin as was reasonable, so that no large heat gradient would form.

Cracking in the first iteration of the roller

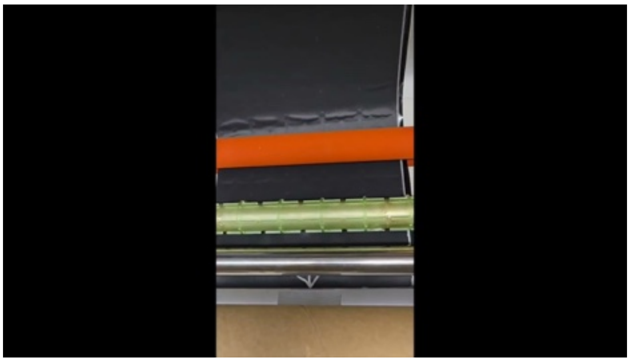
Rotational alignment: The top roller was not driven and relied on friction to turn, and sometimes would slip relative to the bottom roller. Alignment was important to ensure consistent placement of stamps on the fabric, so I put resin gears on the outer edges of the top and bottom rollers so that they would rotate in sync. (Pictured right)



Uneven pressure distribution: I noticed that some of our pouches were coming unsealed at lower than expected pressures, so I ran a single sheet through and looked at what came out. I noted that some places weren’t getting adequate contact/pressure judging by the appearance of the sheet (pictured down right), so I tried using one rubber roller, instead of two resin ones, with the idea that the compliance of the rubber would give more flexibility on how far the profiles stuck out. I also turned down the resin roller in a lathe to ensure a consistent radius. We minimised the amount of material removed while ensuring consistent profile height by putting Sharpie on all the profiles, and turning down with very small passes until all of the black marker was removed.



The spool holders were make mostly of 3D printed parts, along with a cardboard roller and brass rods. Tensioning of the fabric was adjustable by way of a tightenable clamp on the ends of the leading spools, which would vary the torque required to make the spools rotate. The pouch motor fabric was wound up on the upper spool by connecting the drive gears to the spool with a belt and pulley sized to match surface speeds with the stamped roller.



Video demo of the roller machine; watch on YouTube for full screen

3D printer to 3-axis CNC mill conversion

I converted a 3D printer into a CNC machine, using 3D prints made on that same printer. The conversion consisted of switching the belts for lead screws, remounting the stepper motors to accommodate the lead screws, replacing the hotend with a spindle, mounting some spoil board to put stock on, tweaking the firmware, and installing a post processor for Autodesk Fusion's gcode output. The machine worked great for wood and plastic (delrin), and after some adjustment of stepdown, engagement, feeds, and speeds, I got the it working with aluminium, which is great considering how weak and inexpensive the components of this machine are. Until this point I had no experience with CAM, so making this machine was a great opportunity for me to get some experience designing with 3-axis CNC milling in mind, and for gaining an intuition for the many parameters involved when running a CNC machine.

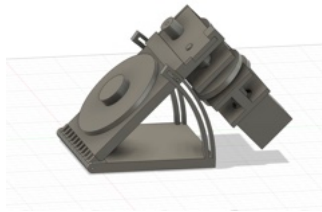
Video of the CNC machine in use

My first aluminium part (pictured above) came loose before the milling operation completed, so it is unfinished, but it still gives us a good idea of the machine's performance, and I will be making the part in full in early September. More tool path optimisation would be very helpful; the machine is rarely working near capacity. Tolerances were within 0.025", which is pretty good considering the runout of the inexpensive spindle

I could use this machine to make a stronger CNC machine, now that I can mill plastics and metals much stronger than 3D prints. Maybe milling steel is next?

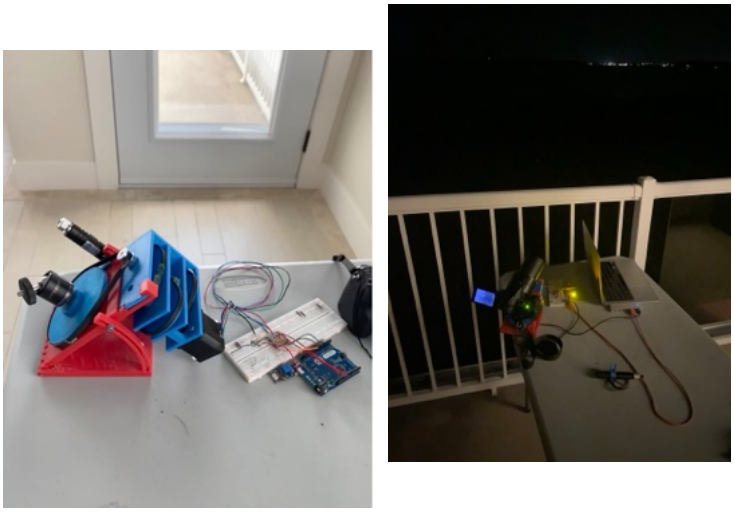
Star Tracker

A star tracker turns a camera to negate the earth's rotation, adding clarity to photos. The device needs to rotate the camera along a parallel axis to the earth's, and at the same speed as the earth's rotation, with great precision. My star tracker achieves this at a fraction of the price of the cost of a commercial star tracker, with similar performance (70 vs 400 USD).

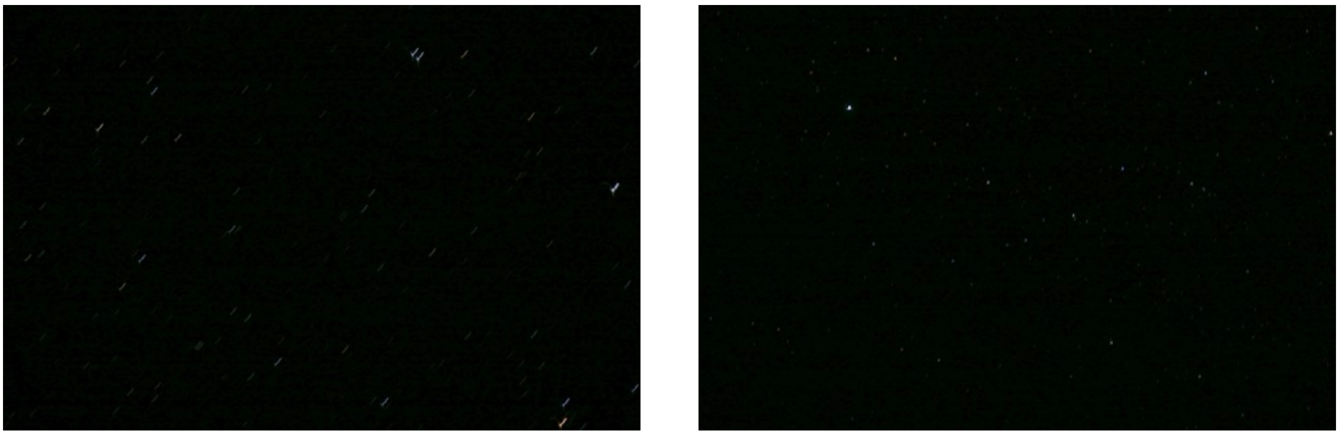


CAD design in Fusion 360

I used Fusion 360 for the 3D printed parts. They constituted the majority of the parts, such as the frame, shafts, and pulleys. Screws were fastened into the plastic parts with heat set inserts. I made an emphasis on having the assembled product have very little slop, because it needs accuracy to produce a good image. This involved incorporating adjustability into features with large tolerances, and using bearings for both radial and axial loads on the final turntable. I used a stepper motor and Arduino to drive the set of belts.



The star tracker with and without a camera mounted.

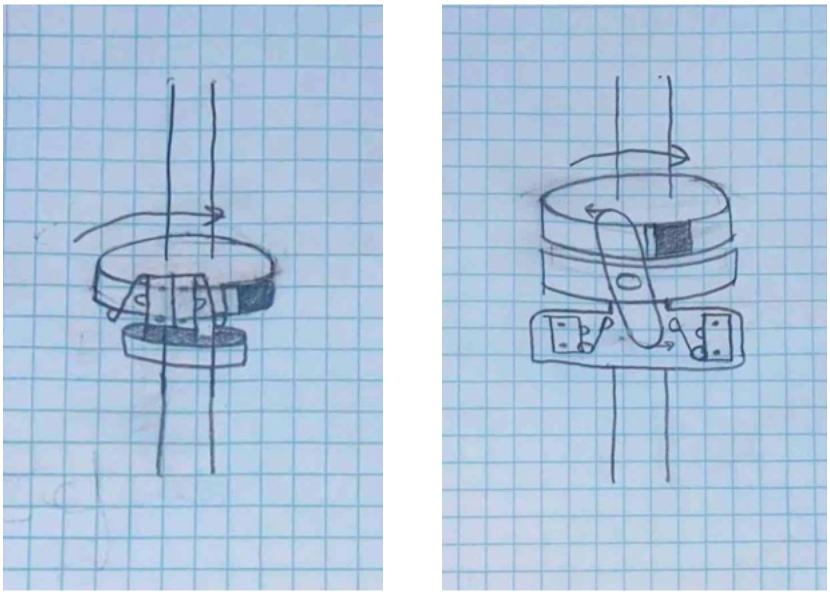


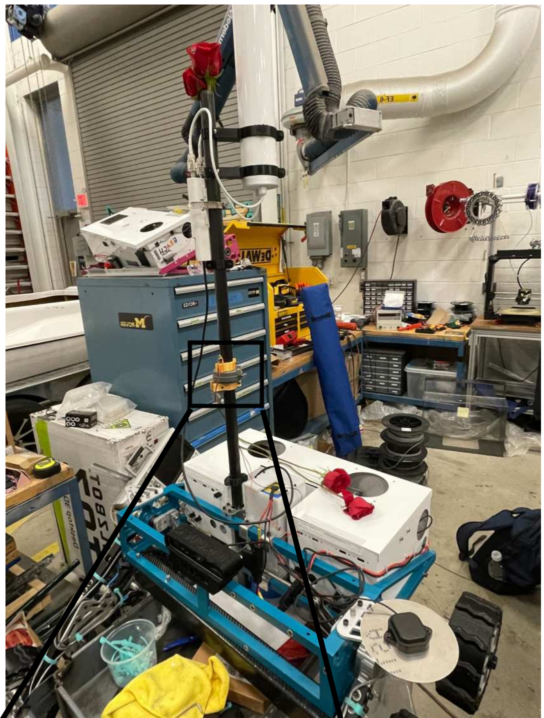
Photos before and after using the star tracker. Notice the streakiness of the stars in the left photo, and their appearance as points on the right.

Timelapse video of star tracker in action: https://youtu.be/KiJlofM_mP8

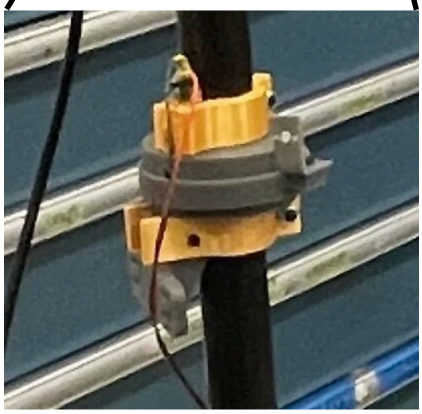
Camera Gimbal Limit Switch Mechanism

The MRover project team Rover has a camera mounted by a two axis gimbal, and to know the exact heading of the camera, the gimbal needs a limit switch at both ends of its range of motion to define start and end angles. In the past, we have done this by having a plastic tab on the rotating part of the gimbal come into contact with the limit switch. This has limited the range of motion of the camera, because the plastic tab and the switches themselves have stopped the rotating piece from making a full 360 degree turn. I developed a mechanism to account for the thickness of the pieces, letting the gimbal rotate all the way around. Instead of the limit switches being mounted to the side of the tab, they will go below it, and be flipped by a two way switch. The two way switch lets the gimbal rotate more than 360 degrees, while still actuating the two limit switches at the end of the gimbal's range of motion. The limit switches will be mounted on the slots visible in the CAD above. The integration of the new system is a work in progress, and as such there are no photos of the final result, with limit switches installed.





Sketch of the gimbal before and after the redesign. Notice that the limit switches occupy space where the plastic tab (shaded rectangle) would otherwise go, and that the two way switch (oval) rotates out of the way of the tab to let it move further.

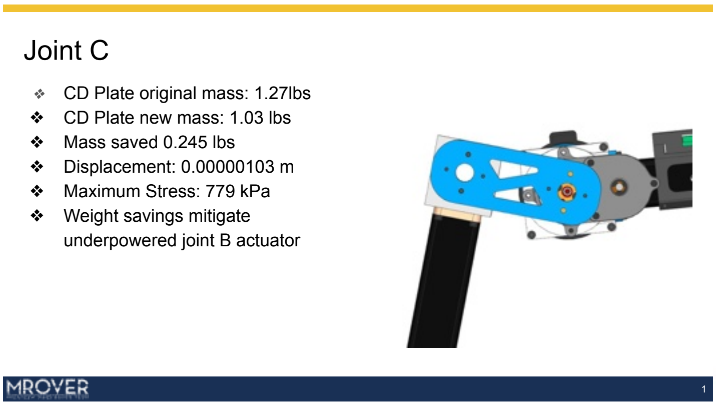


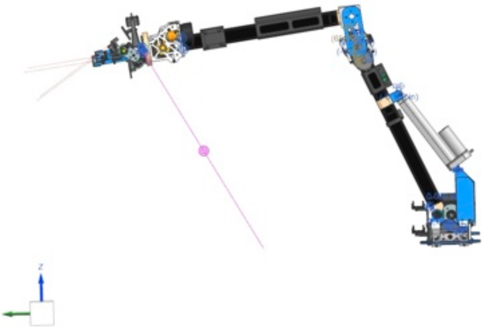
Camera mounted on the gimbal.

Closeup view of the gimbal relative to the entire rover.

Robotic Arm Weight Optimization

I used Ansys to reduce the weight of a plate connecting two arm segments. The linear actuator moving the arm was underpowered, not quite moving within its range of motion within the timeframe set in the University Rover Challenge rules. Reducing this weight reduced the load on the actuator, decreasing the time for the actuator to move through its ROM by up to 5%, depending on the load on the end effector.





End Effector Preliminary Redesign

I redesigned the end effector of the robotic arm to prevent binding of the gripper caused by friction in the lead screw mechanism. Switching to a rack and pinion design reduces static friction in the actuator.